Transcoder-Guided Outlier Selection for Low-Bit LLM Quantization

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Abstract

Low-bit post-training quantization (PTQ) compresses large language models but often destroys task-specific capabilities when only a tiny fraction of weights can remain in full precision. Task Circuit Quantization (TaCQ) selects FP16 outlier weights using contrastive gradients on cross-entropy loss, treating the model as a black box. We propose T-DSO (Transcoder-Guided Dynamic Sparse Overlay), which attributes quantization sensitivity through task-relevant transcoder features (F_{task}) and intersects this signal with standard CE saliency (v2 mult). Under a fixed 0.35% outlier budget matched to TaCQ, T-DSO v2 mult improves Spider execution accuracy on Llama-3.1-8B by +3.4 points on average over four calibration seeds (29.4% vs. TaCQ 26.0%), with seed-to-seed spread 26.2–32.6% (§9.6). Extensions enable dynamic multi-mask execution (§11).

1 Introduction

Large language models (LLMs) are increasingly deployed under strict memory and latency budgets. Post-training quantization (PTQ) maps FP16 weights to 2–4 bits without retraining, but aggressive quantization—especially at 2 bits—can collapse performance on structured tasks such as text-to-SQL, math reasoning, and code generation. Mixed-precision PTQ mitigates this by keeping a small set of outlier weights in FP16 while quantizing the rest [Dettmers et al., 2022, Frantar et al., 2023, Dettmers et al., 2024].

The core problem. Outlier selection is a localization problem: which weights matter for a downstream task once quantization noise is injected? Uniform salience heuristics (magnitude, activation scale) ignore task structure. Calibration on generic corpora misses task-specific circuits. Even strong mixed-precision methods struggle in the 2-bit regime, where the allowed FP16 budget (often <1% of weights) must be spent with surgical precision.

Polysemantic weights vs. monosemantic features. A separate challenge is where in the network task signal lives. Individual MLP neurons and their connecting weights are typically polysemantic—each participates in many unrelated computations via superposition [Elhage et al., 2021, Bricken et al., 2023]. A gradient on cross-entropy therefore reflects a mixture of all roles a weight plays across the vocabulary and prompt positions, not a single task-specific mechanism. TaCQ inherits this ambiguity: although it conditions on task calibration data, saliency is still computed directly on physical weights from CE gradients, so the selected outliers may protect broadly useful directions rather than the sparse features that implement SQL grounding, aggregation, or schema linking.

Sparse dictionary learning over activations—sparse autoencoders and transcoders—pushes representations toward monosemanticity: each latent feature tends to fire for a narrower, more interpretable concept [Bricken et al., 2023, Cunningham et al., 2024, Dunefsky et al., 2024]. This does not eliminate superposition in the base model, but it provides a basis in which task-relevant computation can be isolated before asking which weights must stay in FP16.

Task Circuit Quantization (TaCQ). Xiao et al. [2025] frame outlier selection as automated circuit discovery for PTQ. They score each weight by combining (i) contrastive gradients on task calibration data, (ii) expected weight change under quantization $\Delta W = W - W_{\text{quant}}$, and (iii) a global top-p mask under a fixed memory budget. On Llama-3-8B-Instruct, TaCQ recovers substantial Spider text-to-SQL performance at 2–3 bits when conditioned on task data—e.g., 22.3% vs. 21.9% execution accuracy at 2-bit in our replication ([TODO: add 3-bit and unquantized numbers from Llama-3.1 baseline once available]). Crucially, TaCQ operates in weight space on polysemantic gradient signal; it does not decompose MLP computation into features before ranking outliers.

Why we tackle this differently. Beyond CE aggregating over the full vocabulary, the polysemanticity of base weights means that two weights with similar gradient magnitude may support entirely different latent features—only one of which matters for the task. T-DSO therefore shifts attribution from polysemantic weight gradients to monosemantic feature directions: on clean vs. corrupted contrastive prompts, we identify transcoder features that activate preferentially on task-faithful inputs, reconstruct their MLP contribution, and backpropagate a targeted objective that preserves those feature directions. Physical weight saliency then follows the TaCQ recipe, $S = (|W| \cdot |\nabla_W|) \cdot |W - W_{2\text{bit}}|$, but ∇_W is computed with respect to a feature-filtered loss rather than undifferentiated CE. Under the same outlier budget as TaCQ, we test whether routing saliency through this feature-filtered objective localizes quantization error more precisely than polysemantic CE attribution alone. Our interpretability evidence in §9 combines (i) weight-circuit overlap and causal knockout and (ii) static transcoder feature audits (Appendix A.3)—not full runtime feature tracing, which we defer to the agentic extension.

Our approach (T-DSO). We replace undifferentiated CE attribution with transcoder-directed loss (§5). The pipeline keeps TaCQ's mixed-precision GPTQ stack; only the saliency stage changes.

Contributions.

- We introduce T-DSO, which routes PTQ saliency through sparse transcoder features before mapping to physical weights, intersected with CE saliency in v2 mult.
- We articulate why feature-level attribution departs from TaCQ: same contrastive calibration and ΔW term, but gradients filtered through task-discriminative transcoder latents (§5).
- We provide a fair head-to-head protocol on Llama-3.1-8B-Instruct: identical quantizer (GPTQ), outlier fraction (0.35%), calibration size (128 pairs), and evaluation suites (Spider primary; GSM8K and MMLU in-domain in Tables 8–9).
- On Spider 2-bit, v2 mult improves execution accuracy by +2.9 points on average over four paired seeds (29.4% vs. TaCQ 26.5% $\pm$ 2.0; Table 4), with interpretability analysis showing low mask overlap (J=0.30), causal knockout (-29 pt), and transcoder-native feature audits (§9).
- We release code, contrastive calibration sets, and reproduction scripts for TaCQ and T-DSO masks.[TODO: anonymize for submission]

Paper roadmap. Section 3 states our research questions; Section 4 situates T-DSO relative to PTQ and interpretability; Section 5 describes the algorithm; Sections 6–7 cover setup and findings.

2 Theoretical Framing: Encoded–Used Gap Dissociation

We frame T-DSO against TaCQ as a targeted scalpel vs. a global broadsword. Both preserve a tiny FP16 outlier budget under mixed-precision GPTQ; they differ in what signal selects those outliers.

The encoded–used ambiguity in TaCQ. Cross-entropy on final logits conflates two roles of weights in text-to-SQL:

- Encoding: weights that store schema syntax, SQL templates, and lexical priors (often active whenever SQL-like tokens appear).
- Using: weights on causal execution edges—routing that binds columns, applies aggregations, and composes structure for the current question.

Because CE gradients flow through the full vocabulary, TaCQ's contrastive saliency still mixes encoding and use under a single polysemantic ranking [Xiao et al., 2025]. At 0.35% outliers, wasting budget on passive encoders directly costs execution accuracy on Spider.

Dissociation via F_{task}. T-DSO selects transcoder features with $\Delta a = \text{ReLU}(a_{\text{clean}} - a_{\text{corr}}) > \tau$ and builds y^{target} only from this task set F_{task}. Gradients therefore align with weights that implement contrastive, task-active MLP directions, not every pathway that lowers CE on SQL surface form. We call this encoded–used gap dissociation: outliers target active routing momentum rather than undifferentiated static memory. [TODO: Cite feature-level audit or layer analysis supporting dissociation.]

What we claim in v1 vs. systems extensions. v1 (this paper): same static mask, same quantizer, same budget as TaCQ; test whether dissociation improves Spider execution accuracy at 2–3 bits. Extensions (outlook, §11): distinct sub-task masks, orthogonality constraints, and KV-cache-preserving hot-swaps—architectural capabilities TaCQ's static monolith cannot express.

3 Research Questions

We structure the paper around the following questions. Each maps to a concrete experiment and an ablation slot ([TODO: mark completed vs. pending]).

RQ1: Monosemantic features vs. polysemantic CE attribution. Does attributing through task-relevant transcoder features identify a better outlier set than contrastive CE gradients on polysemantic weights at the same FP16 budget?
Hypothesis: Feature-level loss concentrates gradient on weights that implement monosemantic task directions, avoiding outliers selected for generic LM roles of the same tensors.
Test: T-DSO v2 mult vs. TaCQ on Spider (primary), GSM8K, HumanEval at 2-bit with 0.35% outliers and matched GPTQ calibration.
Status: Spider head-to-head on Llama-3.1-8B: v2 mult mean 29.4 ± 2.7 vs. TaCQ 26.0 seed0 (4-seed paired table: Table 4). GSM8K + three MMLU splits: task-conditioned only (never Spider-calibrated; Tables 8–9).

RQ2: Feature–weight alignment under superposition. Do task-discriminative transcoder features map to distinct weight subsets than CE saliency, reflecting polysemantic vs. monosemantic localization?
Hypothesis: TaCQ and T-DSO masks overlap partially but not fully; low overlap with high T-DSO gain would support the polysemanticity hypothesis.
Test: Weight-level Jaccard overlap; module/layer breakdown (§9.1); static transcoder feature audit (Appendix A.3).
Status: J=0.298 (2-bit) / 0.331 (3-bit); mid layers 4–14 and MLP projections diverge most (Figures 3–4, Table 13); cross-seed stability J=0.995–0.997 (v2) vs. 0.983–0.985 (TaCQ; Table 18, Figure 2); feature Top-K audit script shipped, manual labels pending.

RQ3: Fairness of ΔW. How sensitive is T-DSO to the reference quantized weights in the saliency term $|W - W_{\text{quant}}|$?
Hypothesis: Faithful GPTQ ΔW aligns saliency with actual quantization error; RTN proxy is faster but biased.
Test: Ablate RTN vs. GPTQ corrupt model ([TODO: optional appendix table]). Our main runs use GPTQ 2-bit corrupt weights for both methods.

RQ4: Budget scaling. At what outlier fraction does T-DSO's advantage over TaCQ emerge or vanish?
Test: Sweep mask fraction $\in \{0.15\%, 0.35\%, 0.70\%, 1.5\%\}$ [TODO: future work or appendix].

RQ5: Calibration source (TaCQ Table 4 analogue). How much does the calibration distribution matter at 2–3 bit on MMLU?
Test: Fix eval on MMLU humanities (last 25% of test); vary calib source (WikiText2 vs. STEM vs. matched humanities) for TaCQ and T-DSO (Table 10).
Status: [TODO: pending conditioning-ablation jobs].

RQ7: Scale on dictionary-compatible models. Does v2 mult retain its advantage over TaCQ on larger instruct models where public feature dictionaries exist?
Hypothesis: The CE ∩ feature-filter combiner is architecture-agnostic; relative gain vs. TaCQ persists at 70B even if absolute Spider exec shifts.
Test: Phase 0–2 roadmap (Table 2): CRV 8B (main), Goodfire 8B ablation, Goodfire 70B Spider + GSM8K/MMLU vs. matched TaCQ.
Status: Phase 0 in flight; Phase 2 (70B) [TODO: queued post–8B ledger close].

RQ6: Baseline reproduction (TaCQ on Llama-3.1). Does our ported TaCQ pipeline on Llama-3.1-8B behave consistently with published Llama-3-8B numbers, establishing a fair modern baseline?
Status: Llama-3-8B Spider replication matches paper targets (67.7 / 22.3 / 60.1 exec%); Llama-3.1 full pipeline [TODO: in queue / pending].

4 Related Work

Post-training quantization. GPTQ [Frantar et al., 2023] remains the workhorse for weight-only PTQ of LLMs. Activation-aware methods (AWQ [Lin et al., 2024]), omnidirectional calibration (OmniQuant [Shao et al., 2024]), and 2-bit specialized schemes (QuIP# [Chee et al., 2024]) improve uniform quantization but still degrade on hard reasoning tasks at 2 bits. Mixed-precision and sparse formats—LLM.int8() [Dettmers et al., 2022], SPQR [Dettmers et al., 2024], SliM-LLM [Huang et al., 2025]—retain high-magnitude or salient weights in higher precision via magnitude, Hessian, or group-wise bit allocation. ReALLM [Bedin et al., 2025] compresses each weight matrix with a residual autoencoder plus vector quantization—a different architectural decomposition, not task-conditioned outlier selection. These methods generally define salience without an explicit task-circuit or monosemantic feature basis.

Task-conditioned mixed precision. TaCQ [Xiao et al., 2025] is the closest prior work to ours. Presented at COLM 2025, it uses contrastive calibration, quantization-aware localization (QAL) and magnitude-sharpened gradients (MSG) with a global top-p mask, explicitly connecting PTQ to automated circuit discovery and knowledge localization [Conmy et al., 2023]. Across QA, math, and text-to-SQL on Llama-3 and Qwen2.5, TaCQ outperforms SPQR and SliM-LLM at matched bit budgets—especially at 2–3 bits. We treat TaCQ as the primary baseline because it shares our GPTQ stack, ΔW corrupt model, outlier fraction (0.35%), and Spider evaluation protocol. However, TaCQ scores polysemantic weights via CE on logits; T-DSO intersects CE with saliency routed through a monosemantic transcoder basis (§5). Empirically, our v2 mult mask overlaps only 30% (Jaccard J=0.298) with TaCQ at the same budget (§9.1), showing the gain is not a minor reranking within TaCQ's weight pool.

Superposition, polysemanticity, and sparse features. Elhage et al. [2021] argue that capacity constraints force LLMs to encode more features than neurons, yielding polysemantic units whose activations mix many concepts. Dictionary learning and sparse autoencoders (SAEs) recover latents closer to monosemanticity—each feature aligns with a narrower behavior [Bricken et al., 2023, Cunningham et al., 2024]. Transcoders apply the same idea at MLP boundaries: they approximate MLP(x) with a wider, sparsely activating layer [Dunefsky et al., 2024], enabling cleaner circuit factorization than residual-stream SAEs [Wright et al., 2025]. Prior work uses these tools for analysis, steering, or verification (§4); we ask whether monosemantic feature attribution should drive compression by deciding which polysemantic weights must remain in FP16.

Mechanistic interpretability, transcoders, and circuit verification. Transformer circuits [Elhage et al., 2021] and automated circuit discovery [Conmy et al., 2023] seek sparse subgraphs explaining behaviors. Meta's Circuit Tracer and the public crv-8b-instruct-transcoders weights provide per-layer TopK transcoders for Llama-3.1-8B-Instruct [Ameisen et al., 2026]. Circuit-based Reasoning Verification (CRV) [Ameisen et al., 2026] replaces MLPs with these transcoders to build attribution graphs, classify reasoning errors, and intervene on individual features—the same sparse basis we use, but for runtime verification rather than offline weight selection. Unlike TaCQ, which ranks polysemantic weights from CE gradients, T-DSO attributes through this monosemantic feature basis first, then maps saliency back to physical parameters.

Table 1: Positioning vs. closest prior work (all target low-bit LLM compression or control).

Method	Goal	Feature basis	Weights?
SPQR / AWQ / SliM-LLM	PTQ	Magnitude / Hessian / salience	✓
TaCQ [Xiao et al., 2025]	Task PTQ	CE (polysemantic)	✓
SAS / SAE-TS [Bayat et al., 2025, Chalnev et al., 2024]	Steer	SAE latents	—
CRV [Ameisen et al., 2026]	Verify CoT	Transcoder latents	—
T-DSO (ours)	Task PTQ	Transcoder ∩ CE	✓

Sparse-feature steering and contrastive attribution. Contrastive prompts (clean vs. corrupted inputs) appear in influence estimation, data attribution, and TaCQ's gradient capture. Sparse Activation Steering (SAS) [Bayat et al., 2025], also at COLM 2025, uses contrastive prompt pairs to isolate behavior-specific SAE latents and steer model activations at inference— sharing our motivation to escape superposition, but modifying activations at test time rather than selecting weights at compression time. SAE-Targeted Steering [Chalnev et al., 2024] constructs steering vectors by modeling effects on SAE features; again, the output is a steering direction, not a quantization mask. We reuse contrastive calibration infrastructure but change what is being attributed: transcoder feature activations on MLP hooks (and their intersection with CE), combined with $|W| \cdot |\Delta W|$ as in TaCQ.

Quantization vs. interpretability. Recent work studies how PTQ damages SAE features even when perplexity looks stable [Anonymous, 2026], motivating feature-level audits of compressed models. Our direction is complementary: we use interpretability tools upstream to choose a mask that preserves task performance, then measure whether the selected circuit differs from CE-only selection.

Text-to-SQL as a testbed. Spider [Yu et al., 2018] is a standard semantic parsing benchmark requiring schema grounding and compositional SQL. It stress-tests structured output under quantization and matches TaCQ's headline results. We extend evaluation to [TODO: GSM8K, HumanEval] for generality.

Positioning. Table 1 summarizes how T-DSO relates to the closest lines of work. To our knowledge, no prior conference paper uses transcoder (or SAE) task-feature attribution to select PTQ outlier weights under a fixed FP16 budget and reports both (i) downstream task gains vs. TaCQ and (ii) low mask overlap showing a distinct circuit. T-DSO sits at the intersection of task-aware PTQ [Xiao et al., 2025, Huang et al., 2025] and monosemantic feature attribution [Dunefsky et al., 2024, Bayat et al., 2025, Ameisen et al., 2026]. Unlike TaCQ, we do not treat weights as atomic units of task importance—we decompose MLP computation to mitigate polysemanticity before selecting outliers. Unlike steering and verification work, we do not seek human-readable explanations or runtime control for their own sake—we convert feature-level task signal into a weight mask with measurable downstream impact (+3.4 pt mean Spider exec at 2-bit vs. TaCQ, four seeds; §9.6).

5 Method: T-DSO

We keep the TaCQ mixed-precision quantization pipeline (GPTQ with per-weight bit-width mask) and replace only the saliency extraction stage.

Setup. Model f_θ with weights θ; uniform 2-bit reference $W_{2\text{bit}}$ from GPTQ on task calibration data; outlier budget k (fraction of parameters kept in FP16). Contrastive pairs (x^+, x^-) where x^+ is task-faithful and x^- is corrupted (schema errors, wrong SQL, etc.).

TaCQ saliency (review). Accumulate $|\nabla_W \mathcal{L}_{\text{CE}}(x^+)|$ over calibration, multiply by $|W| \cdot |W - W_{2\text{bit}}|$, global top-$k$ selection [Xiao et al., 2025]. Gradients are taken w.r.t. polysemantic weight tensors: each W may contribute to many superposed features, and $\mathcal{L}_{\text{CE}}$ mixes all roles active at each token.

From polysemantic weights to monosemantic features. T-DSO inserts a feature-space bottleneck before weight saliency. For each layer ℓ, transcoder $\mathcal{T}_\ell$ maps MLP input to a sparse code $f \in \mathbb{R}^F_\ell$ with $F_\ell \gg d_{\text{mlp}}$ but only TopK entries nonzero per token—a dictionary whose entries are approximately monosemantic [Bricken et al., 2023, Cunningham et al., 2024]. We never rank features globally for interpretability alone; we rank task-discriminative features via contrastive activation: $a^{\text{corr}}_{\ell,f} = \max_t f_{\ell,t,f}$. On clean input x^+ (with grad), compute $a^{\text{clean}}_{\ell,f}$ analogously. Task features satisfy $\Delta a_\ell = \text{ReLU}(a^{\text{clean}}_\ell - a^{\text{corr}}_\ell) > \tau_\ell$ (threshold τ_ℓ = per-example quantile, default 95th percentile of positive Δa). Target MLP output contribution: $y^{\text{target}}_\ell = \mathcal{T}_\ell.\text{decode}(f^{\text{clean}}_\ell \odot \mathbf{1}_{\Delta a > \tau})$. Loss per layer: $\mathcal{L}_\ell = -\sum_t \langle h^{\text{mlp,out}}_{\ell,t}, y^{\text{target}}_{\ell,t} \rangle$. Total $\mathcal{L} = \sum_\ell \mathcal{L}_\ell$; one backward pass accumulates $|\nabla_W \mathcal{L}|$ (sample_abs semantics, matching TaCQ). Only weights on paths that implement the selected monosemantic features receive gradient; polysemantic CE would additionally reward weights tied to non-task features active at the same tokens.

Why this differs from TaCQ. Both methods share contrastive pairs, ΔW, and global top-k over physical weights. The distinction is attribution basis: TaCQ asks "which polysemantic weights does CE care about?"; T-DSO asks "which weights implement monosemantic features that distinguish clean from corrupt task inputs?". [TODO: Figure: polysemantic weight gradient vs. feature-filtered path to weights.]

Physical weight saliency and mask. For each weight tensor W, define per-weight CE saliency g_{CE} as in TaCQ and transcoder saliency g_{align} from $\mathcal{L}$ above (same $|W|$ and $|W - W_{2\text{bit}}|$ scaling). Our best variant (v2 mult) combines both via geometric mean:

$$S(W) = |W| \cdot |W - W_{2\text{bit}}| \cdot \sqrt{g_{\text{CE}}(W)\, g_{\text{align}}(W)} \tag{1}$$

selecting weights salient under both polysemantic CE and monosemantic feature routing. Replacing g_{CE} with g_{align} alone (v1) ties TaCQ under fair calibration; union-style boosts ($g_{\text{CE}}(1 + \lambda g_{\text{align}})$) degenerate at 2-bit. Global top-k over all eligible linear weights yields binary mask M; GPTQ quantizes $M(W)$=0 to 2-bit (or 3-bit) and keeps $M(W)$=1 in FP16.

5.1 Anchored union: consensus core and dispute region

Empirically, TaCQ and v2 mult agree on only ∼46–50% of the FP16 budget (J_{Jaccard}≈0.30–0.33; §9.1), yet both methods rank weights with the same global top-k operator applied to different saliency fields. We formalize a two-region policy that never renegotiates consensus outliers and allocates only the remaining budget on the method-disputed mass.

Setup. Let Ω denote the set of eligible weight indices (all attention and MLP projections). Write $s_{TaCQ}(w)$ and $s_{T-DSO}(w)$ for the final per-weight saliency scores of TaCQ and v2 mult respectively (Eq. 1 with the appropriate gradient term for each method), and k for the global FP16 budget ($k = \lfloor \rho|\Omega| \rfloor$, ρ=0.0035). Each method induces an independent mask by global top-k:

$$M_{TaCQ} = \text{TopK}\big(\{s_{TaCQ}(w)\}_{w\in\Omega},\, k\big), \tag{2}$$

$$M_{T-DSO} = \text{TopK}\big(\{s_{T-DSO}(w)\}_{w\in\Omega},\, k\big). \tag{3}$$

Decomposition. Define the consensus core and dispute region:

$$M_{\text{core}} \triangleq M_{TaCQ} \cap M_{T-DSO}, \tag{4}$$

$$M_{\text{dispute}} \triangleq M_{TaCQ} \triangle M_{T-DSO} \;=\; \big(M_{TaCQ} \setminus M_{T-DSO}\big) \cup \big(M_{T-DSO} \setminus M_{TaCQ}\big). \tag{5}$$

Weights in M_{core} are FP16 under both saliency fields; weights in M_{dispute} appear in exactly one method's mask. In our Spider runs $k_{\text{core}} \triangleq |M_{\text{core}}| \approx 1.12 \times 10^7$ (∼46% of k) and $|M_{\text{dispute}}| \approx 2.64 \times 10^7$, with $k_{\text{fill}} \triangleq k - k_{\text{core}} \approx 1.32 \times 10^7$ slots to assign inside the dispute region.

Two-stage selection (v3). Stage 1 locks the core: $M \supseteq M_{\text{core}}$ unconditionally. Stage 2 fills the residual budget k_{fill} only from M_{dispute} using a different ranking rule than either method used globally. Let $\widehat{s}_{T-DSO}(w) = s_{T-DSO}(w)/(\max_{w'} s_{T-DSO}(w') + \epsilon)$ and $\widehat{s}_{TaCQ}(w)$ analogously (per-tensor normalization). For $\lambda \in [0, 1]$, define the dispute score

$$s_\lambda(w) \;=\; \lambda\,\widehat{s}_{T-DSO}(w) + (1-\lambda)\,\widehat{s}_{TaCQ}(w), \qquad w \in M_{\text{dispute}}, \tag{6}$$

and zero elsewhere. The anchored-union mask is

$$M_\lambda \;=\; M_{\text{core}} \,\cup\, \mathrm{TopK}\big(\{s_\lambda(w)\}_{w\in M_{\text{dispute}}},\, k_{\text{fill}}\big). \tag{7}$$

Endpoints: λ=0 ranks dispute mass by TaCQ (MLP-heavy globally); λ=1 by T-DSO (attention-heavy); λ=0.5 blends both fields on the disputed mass only. GPTQ then proceeds unchanged with mask M_λ.

Relation to v2 mult. v2 mult is a single-stage rule: one saliency field $s_{T-DSO} \propto \sqrt{g_{\text{CE}}\, g_{\text{align}}}$ followed by one global top-k. Anchored union is a two-stage rule on the pair (M_{TaCQ}, M_{T-DSO}): consensus is inherited from both rankings; disagreement is resolved by a controlled interpolation (Eq. 6–7). Thus v2 mult is not required to equal any M_λ, but the sweep over λ tests where the optimal dispute allocation lies as quantization severity changes (Figure 6, Tables 14–15).

Design rationale. (i) M_{core} is a method-agnostic lower bound on critical mass: any weight both CE and feature routing insist on keeping is never demoted. (ii) Only $|M_{\text{dispute}}|$ is re-ranked, so the experiment isolates the routing-vs-memory trade-off identified in §9.1 without perturbing shared outliers. (iii) Bit-width enters only through $W_{2\text{bit}}$ (or $W_{3\text{bit}}$) in $s(\cdot)$ and through GPTQ noise; λ can be chosen per bit-width ($\lambda^*(b)$) if exec peaks shift across regimes.

Implementation notes. Transcoders stream from CPU to avoid VRAM blow-up; saliency ranks on CPU. Model: Llama-3.1-8B-Instruct (transcoder-compatible). [TODO: Add pseudocode Algorithm 1; figure showing clean/corrupt/transcoder path.]

Complexity. Same asymptotic cost as TaCQ (one forward-backward per calibration example) plus transcoder encode/decode per layer per forward. [TODO: Report wall-clock vs. TaCQ on 128 pairs.]

6 Experimental Setup

Model and interpretability stack. Llama-3.1-8B-Instruct via unsloth mirror (bit-identical to Meta weights). Transcoders: facebook/crv-8b-instruct-transcoders (32 layers, TopK, F=131,072). All head-to-head comparisons use this model; Llama-3-8B TaCQ numbers reproduce the original paper on a separate track.

Tasks and metrics (TaCQ paper protocol).

- Spider text-to-SQL: execution accuracy on dev (1034 examples, test-suite-sql-eval).
- GSM8K: 8-shot chain-of-thought; flexible numeric extract (GSM8k_eval.py).
- MMLU: 5-shot MCQA; three subject splits (MMLU_STEM, MMLU_humanities, MMLU_social_sciences), each conditioned separately [Xiao et al., 2025]; calib on first 75% of each subject test CSV, eval on last 25% (eval_start_p=0.75).
- HumanEval: not bundled in the upstream TaCQ codebase; deferred.

Calibration (task-conditioned, TaCQ protocol). Corrupt model, importances, mask, and masked GPTQ are calibrated on the same task as evaluation (paper/TACQ_FAIR_PROTOCOL.md). GSM8K and MMLU models are never Spider-conditioned.

- Spider — train split (1360 examples).
- GSM8K — train split (8-shot CoT prompts); eval on test.
- MMLU — 5-shot from dev; calib on first 75% of each subject test file; eval on last 25%.

1360 contrastive pairs from Spider train for Spider head-to-head only. Per-task GSM8K/MMLU TaCQ replicates use native measure_importances.

Multi-seed protocol (paired robustness). We report four seeds ($s \in \{0, 1, 2, 3\}$) for both TaCQ and v2 mult (Table 4). All seeds share the same GPTQ 2-bit corrupt ΔW (seed0 checkpoint). TaCQ sets serial_number=s in measure_importances (128-example Spider-train shuffle + gradient/GPTQ RNG) and –seed s in gptq.llama. v2 mult uses seed-specific contrastive JSONL (1360 train pairs) and extract_tdso_v2 –seed s. Scripts: run_tacq_multiseed_validate.sh, run_tdso_v2_validate.sh, summarize_paired_seeds.sh.

Quantization protocol (matched for fairness).

- Outlier fraction: 0.35% (mask_fraction=0.0035), proportional over 224 attn+MLP weight tensors (–proportional_total_params).
- Reference ΔW: GPTQ 2-bit corrupt model (128 samples, true-sequential).
- Final quant: GPTQ with fine-wbits-yaml from outlier mask; ranking top_p_sparse.
- Bit widths: 2-bit and 3-bit [TODO: report both].

Baselines.

- FP16: unquantized instruct model.
- Uniform GPTQ: no outlier mask.
- TaCQ: contrastive CE saliency [Xiao et al., 2025].
- T-DSO (ours): transcoder-directed saliency (§5).

[TODO: SPQR / AWQ optional extended table.]

Hardware and reproducibility. 4× H200/GH200 GPUs per node; full replication scripts under scripts/ and TACQ/. Llama-3.1 environment: llama31_venv (transformers 4.45+). [TODO: Report total GPU-hours.]

Replication sanity check (Llama-3-8B). Our harness reproduces TaCQ Spider exec accuracy: 67.7 (FP16), 22.3 (2-bit), 60.1 (3-bit) vs. paper 67.6 / 21.92 / 58.32.

6.1 Scale evaluation (transcoder-compatible models)

Primary results use Llama-3.1-8B-Instruct with Meta CRV TopK MLP transcoders (facebook/crv-8b-instruct-transcoders)—currently the only public full-stack per-layer transcoder dictionary for an instruct LLM. We commit to extending T-DSO to scaled models wherever a compatible dictionary exists, using the same matched GPTQ protocol (0.35% outliers, task calibration, v2 mult).

Phase 0 (8B, submission-critical). Spider head-to-head, GSM8K + MMLU secondary tasks, multi-seed robustness, and mechanism controls (domain-shift, circuit knockout, contrastive squeeze) on CRV–8B.

Table 2: Scale roadmap: models with public interpretability dictionaries. T-DSO vs. TaCQ only where a dictionary column is filled.

Phase	Model	Dictionary	T-DSO vs. TaCQ
0	Llama-3.1-8B	CRV MLP (32L)	✓ (main)
1a	Llama-3.2-1B	EleutherAI skip-TC	✓ (ladder)
1b	Llama-3.1-8B	Goodfire SAE L19	✓ (ablation)
2	Llama-3.3-70B	Goodfire SAE L50	✓ (scale)
3	Qwen2.5-7B / Llama-3-70B	—	TaCQ only

Table 3: Spider execution accuracy (%) on Llama-3.1-8B-Instruct. Outlier budget 0.35% (24.4M weights). Calibration: 1360 Spider train examples. T-DSO v2 mult: mean ± sample std over four calibration seeds (Table 4); TaCQ: matched four-seed sweep (same ΔW).

Method	FP16	2-bit
FP16 (unquantized)	67.8	—
TaCQ [Xiao et al., 2025] (mean)	—	26.5 ± 2.0
TaCQ [Xiao et al., 2025] (seed0)	—	26.0
T-DSO v1 (align-only)	—	25.0
T-DSO v2 mult (mean)	—	29.4 ± 2.7
T-DSO v2 mult (best seed)	—	32.6

Phase 1 (dictionary ladder). EleutherAI skip-transcoders on Llama-3.2-1B (cheap sanity) and Goodfire instruct SAEs on Llama-3.1-8B layer 19 (cross-dictionary ablation vs. CRV at fixed width).

Phase 2 (70B scale headline). Llama-3.3-70B-Instruct with Goodfire/Llama-3.3-70B-Instruct-SAE-l50—public instruct SAE at scale—for Spider + GSM8K/MMLU vs. matched TaCQ MSG. Full-layer CRV transcoders for 70B are not yet released; Goodfire provides the interim scale path with identical contrastive feature-filter logic (§5).

Phase 3 (TaCQ-only baselines). Qwen2.5-7B and Llama-3-70B TaCQ runs (no T-DSO) optional for cross-architecture context; they do not test the transcoder claim.

Table 2 summarizes targets; full runbook in paper/SCALE_PLAN.md.

[TODO: Fill Phase 2 rows when run_scale_70b_goodfire.sh completes.]

7 Results

7.1 Spider text-to-SQL (Llama-3.1-8B-Instruct)

Table 3 reports execution accuracy on the Spider dev set (1034 examples, test-suite-sql-eval) under a matched 0.35% FP16 outlier budget. All mixed-precision models share the same GPTQ 2-bit base, faithful ΔW corrupt checkpoint, 1360 train-split contrastive calibration examples (leak-free), and greedy generation with extract_sql post-processing (§6).

RQ1: v2 mult vs. TaCQ at 2-bit. Table 4 reports a paired four-seed sweep: both methods share the same GPTQ 2-bit corrupt checkpoint and Spider task, but vary calibration stochasticity via seed (§9.6). v2 mult averages 29.4% vs. TaCQ 26.5% ± 2.0 (+2.9 pp mean paired Δ; Table 4). The best v2 seed reaches 32.6% (+6.6 vs. TaCQ s0), but seed 1 (26.2%) ties TaCQ s0, so the headline gain is seed-sensitive for both methods once paired. Replacing CE entirely with transcoder alignment (v1) ties TaCQ under fair calibration (25.0 vs. 26.0), so improvement requires combining task CE with task-circuit attribution—not either signal alone. Per-difficulty breakdown for the best seed (32.6%): easy 68.1, medium 29.1, hard 17.2, extra 4.8 (vs. TaCQ 2-bit: 60.5, 21.5, 12.1, 1.2).

Table 4: Paired multi-seed robustness at 2-bit Spider (leak-free dev eval). TaCQ: full MSG pipeline with serial_number=s (128-example Spider calib shuffle + GPTQ seed s; same ΔW corrupt for all seeds). T-DSO v2 mult: seed-specific contrastive JSONL + extraction –seed. Δ = v2 − TaCQ (percentage points). Mask J: seed-matched v2 vs. TaCQ (J=0.298 at all seeds; ±0.001).

Seed	TaCQ (%)	v2 mult (%)	Δ	Mask J (method)
0	26.0	32.6	+6.6	0.298
1	29.6	26.2	-3.4	0.298
2	25.0	28.6	+3.6	0.298
3	25.5	30.1	+4.6	0.298
Mean ± std	26.5 ± 2.0	29.4 ± 2.7	$+2.9 \pm 3.9$	—

Table 5: Outcome matrix: TaCQ (global broadsword) vs. T-DSO (targeted scalpel). †=v1 empirical; ‡=systems outlook (§11).

Dimension	TaCQ	T-DSO
Mask targeting	$\mathcal{L}_{\mathrm{CE}}$ on logits †	CE ∩ transcoder circuit †
Weight preservation	Polysemantic (mixed) †	Dissociated routing †
Execution topology	Static single-pass †	Dynamic multi-agent overlays ‡
KV across agent steps	Recompute prefill ‡	Inherit cache, swap overlay ‡

RQ2: low overlap explains the gain (seed0 mask). Mask Jaccard between v2 mult and TaCQ is J=0.298 (2-bit) / 0.331 (3-bit); ∼54% of T-DSO outliers are unique at 2-bit (§9.1, Figures 1–4). Cross-seed mask stability differs sharply by method ($J_{\text{v2 cross}}$=0.995–0.997 vs. $J_{\text{v2 vs TaCQ}}$=0.298; Figure 2).

Causal validation (circuit knockout). Independent of PTQ gains, targeted suppression of the T-DSO mask (α=0.8, Eq. 8) drops Spider exec from 67.8% to 38.9% while English probes stay fluent (§9.2). This addresses the standard reviewer concern that outlier masks may be correlated but not causal.

In-memory contrastive squeeze. Mild simultaneous task-boost / background-damp scaling (α_{task}=1.05, α_{base}=0.98), selected on a train holdout only, yields 68.8% dev exec vs. 67.8% contemporaneous FP16 (+1.0 pt; Table 17, §9.3).

Mask-method ablation (2-bit, Spider). Table 6 closes the ledger on causal vs. load-bearing masks (§8). B0 (F_{task}) supports matched ΔW corruption (+3.4 pt mean vs. TaCQ) but fails as a GPTQ outlier map (0.0–18.0% under native gptq.llama). MSG preserves 26.0% only with Spider calibration; Phase B transcoder-stacked GPTQ (B0 mask, $\lambda \in \{0, 0.1, 0.5, 1.0\}$) is flat at 18.0%—not competitive with either mask pipeline.

3-bit. Under the same leak-free protocol, TaCQ 3-bit reaches 64.8%; fair v1 T-DSO ties at 65.0%. v2 mult 3-bit (seed0) reaches 66.2% (+1.4 vs. TaCQ); multi-seed 3-bit sweep is future work. The v3 anchored-union dispute sweep (Section 5.1) shows a complementary pattern at partial coverage: TaCQ-heavy dispute allocation (λ_{bias}=0) reaches 63.2% vs. 59.9% at balanced λ_{bias}=0.5 (Table 15), consistent with milder 3-bit quantization favoring MSG-style MLP retention over routing-heavy T-DSO mass. Uniform no-mask GPTQ 2-bit collapses to 0.0% (Phase B v1 baseline).

Harness note. An early Llama-3.1 baseline (44.3% FP16) used raw split(";") parsing that kept "'sql markdown fences in predictions; corrected extract_sql parsing yields 67.8% FP16, matching the Llama-3-8B paper target. All numbers above use the corrected harness.

Secondary tasks (GSM8K + MMLU). Following Xiao et al. [2025], we report task-conditioned models only: for each benchmark T, corrupt model, saliency, mask, and masked GPTQ are calibrated on T's

Table 6: 2-bit Spider ablation: mask identity $\times$ calibration / quant path (0.35% budget). All rows use gptq.llama except ΔW corrupt (no re-quant).

Mask	Calibration / method	Exec (%)
Load-bearing (MSG) — GPTQ		
MSG	Spider (full TaCQ)	26.0
MSG	Spider (128, native control)	26.0
MSG	C4	0.6
Causal (B0) — GPTQ		
B0 heavy	C4	0.0
B0 heavy	Spider (Phase B stacked, λ=0–1.0)	18.0
Causal (B0) — ΔW corrupt (no GPTQ)		
B0 heavy	Spider ΔW (v2 mult, mean 4 seeds)	29.4

Table 7: Llama-3-8B TaCQ reproduction (sanity check vs. Xiao et al. [2025]).

Setting	Ours	Paper
Unquantized	67.7	67.6
TaCQ 2-bit	22.3	21.92
TaCQ 3-bit	60.1	58.32

calibration split and evaluated on T's held-out split (Tables 8 and 9). Conditioning a Spider-trained mask on GSM8K/MMLU is not part of the TaCQ protocol and is omitted here—calibration distribution dominates low-bit performance ([TODO: Table 10]). Spider remains the primary head-to-head task; GSM8K and the three MMLU splits use task-native calibration only (§6).

Paired multi-seed TaCQ. We re-run the full TaCQ MSG pipeline for seeds 1–3 (run_tacq_multiseed_validate.sh): measure_importances with serial_number=s shuffles which 128 Spider-train examples drive contrastive gradients; GPTQ uses –seed s; corrupt ΔW is fixed (seed0 checkpoint). Cross-seed TaCQ mask Jaccard and paired Δ statistics are logged in tacq_crossseed_jaccard.log and paired_seed_summary.log ([TODO: fill when multiseed 3-bit job completes]).

Cross-task conditioning (RQ5). Following Xiao et al. [2025], we study which calibration distribution is used (Table 10), not Spider→GSM8K mask transfer. Per-task numbers in Tables 8–9 condition on the same split as eval.

8 Domain-shift at 2-bit: calibration corpus controls

Standard GPTQ pipelines often calibrate on generic web text (e.g., C4). We isolate the effect of calibration domain at fixed 2-bit width, mask budget (0.35%), and gptq.llama implementation, using pre-computed masks without re-running saliency.

Table 11 shows a 25+ point collapse when switching MSG from Spider to C4 calibration at 2-bit, while the closing Spider-native control reproduces the full TaCQ 26.0% baseline. Generic corpus calibration is therefore a confound in 2-bit task evaluations; task-matched calibration is necessary even when the outlier mask is held fixed.

Table 8: Task-conditioned 2-bit (0.35% budget): calib and eval on the same task. Mean ± std over three calibration seeds (serial 0–2) when available; seed0 from downstream2 job, seeds 1–2 from downstream_multiseed. GSM8K: 8-shot CoT, train calib / test eval. MMLU: 5-shot from dev; calib first 75% of test per subject, eval last 25%. Never Spider-conditioned.

Task	FP16	TaCQ	T-DSO v2 mult
GSM8K (flex EM)	69.3	[TODO: TBD]	64.9
MMLU STEM	59.0	[TODO: TBD]	[TODO: TBD]
MMLU humanities	66.0	[TODO: TBD]	[TODO: TBD]
MMLU social sciences	75.6	[TODO: TBD]	[TODO: TBD]

Table 9: Task-conditioned 3-bit: same protocol as Table 8.

Task	FP16	TaCQ	T-DSO v2 mult
GSM8K (flex EM)	69.3	[TODO: TBD]	[TODO: TBD]
MMLU STEM	59.0	[TODO: TBD]	[TODO: TBD]
MMLU humanities	66.0	[TODO: TBD]	[TODO: TBD]
MMLU social sciences	75.6	[TODO: TBD]	[TODO: TBD]

9 Analysis

9.1 Mask overlap: a different circuit, not a reranking

Both T-DSO (v2 mult) and TaCQ keep exactly 0.35% of the 224 attn+MLP weight tensors—24.4M FP16 outliers each—yet they select largely disjoint subsets. We compute pairwise weight-level Jaccard on saved mask checkpoints (scripts/analysis/plot_jaccard_figures.py; logs in crossseed_jaccard.log, tacq_crossseed_jaccard.log, jaccard_3bit_cross_method.log). Table 12 reports global overlap; Figures 1–4 visualize cross-seed stability, cross-bitwidth gaps, and layer/module structure.

Against TaCQ 2-bit (seed-matched), only 46% of T-DSO outliers overlap (J=0.298); against TaCQ 3-bit (matched bitwidth) the overlap is 50% (J=0.331). Thus ∼54% of T-DSO's FP16 budget (13.2M weights vs. TaCQ 2-bit) protects weights TaCQ would quantize—this is not a minor reranking within the same circuit.

Module-level divergence. Table 13 and Figure 5 decompose disagreement by projection type. MLP blocks (down_proj, gate_proj, up_proj) contribute the largest TaCQ-exclusive mass (∼2.9–3.8M each). In contrast, T-DSO retains more attention weights—notably q_proj, k_proj, and v_proj, where T-DSO-only counts exceed TaCQ-only by ∼0.4–1.3M per head type. Aggregating attention vs. MLP, T-DSO adds 5.5M routing-centric outliers vs. 3.0M for TaCQ, while TaCQ adds 10.2M MLP-only vs. 7.7M for T-DSO. The transcoder steers the budget toward mid-network routing (attention) while reshuffling which MLP channels stay in FP16.

Layer-wise divergence. Figures 3 and 4 show per-layer decomposition. Disagreement peaks in mid layers 4–14 (layers 8–11 carry ∼1.2M combined exclusive weights each; layer 9 is the single most divergent). Early and late layers agree more; the mid-stack is where task-specific schema and join structure are thought to be processed in Llama-family models [Xiao et al., 2025]. The mean paired exec gain at 2-bit (+2.9 pp; Table 4) therefore tracks a substantially different outlier circuit, consistent with RQ2: transcoder-guided saliency localizes complementary weights rather than filtering the same polysemantic pool.

Interpretation for RQ2. High T-DSO gain together with low Jaccard supports our polysemanticity hypothesis: TaCQ CE gradients and T-DSO feature-filtered gradients prioritize different physical weights even under identical ΔW, quantizer, and budget. The v2 mult combiner (intersection of CE and transcoder signals) outperforms either signal alone (v1 align-only ties TaCQ at 25.0 vs. 26.0), suggesting the winning mask is the intersection of task-sensitive and task-circuit saliency—not a wholesale replacement of CE.

Table 10: Calibration-source ablation (TaCQ Table 4 analogue): MMLU humanities accuracy when conditioning on different calibration sets, 2- vs. 3-bit. Eval fixed on humanities; compare WikiText2 / unrelated MMLU split / matched split.

Calib source	2b TaCQ	2b T-DSO	3b TaCQ	3b T-DSO
WikiText2	[TODO: TBD]	[TODO: TBD]	[TODO: TBD]	[TODO: TBD]
MMLU STEM	[TODO: TBD]	[TODO: TBD]	[TODO: TBD]	[TODO: TBD]
MMLU humanities (matched)	[TODO: TBD]	[TODO: TBD]	[TODO: TBD]	[TODO: TBD]

Table 11: 2-bit Spider dev exec under native gptq.llama (128 samples, true-sequential). Same quantizer; only mask identity and calibration corpus vary.

Outlier mask	GPTQ calibration	Exec (%)
TaCQ MSG	C4	0.6
TaCQ MSG	Spider (128)	26.0
B0 heavy (T-DSO)	C4	0.0
TaCQ full pipeline (Spider contrastive 1360)		26.0

Dispute sweep (v3 anchored union). The anchored-union construction (Section 5.1) locks the consensus core $M_{\text{core}} = M_{TaCQ} \cap M_{T-DSO}$ ($\sim$11.2M weights at 2-bit, 46% of the budget; $\sim$12.1M / 50% at 3-bit) and re-ranks only the XOR dispute mass ($\sim$13.2M fill slots) by blending normalized TaCQ vs. T-DSO saliency with bias λ_{bias} (0 = TaCQ-heavy, 1 = T-DSO-heavy). Table 14 shows the intended continuous mask morph: at λ_{bias}=0 the v3 mask overlaps TaCQ at J=0.708 (2-bit) but only J=0.460 with v2 mult; at λ_{bias}=1 the pattern reverses (J_{TaCQ}=0.462, J_{T-DSO}=0.705). Crossover occurs near λ_{bias}=0.5 ($J_{\text{TaCQ}} \approx J_{T-DSO} \approx 0.57$), confirming the sweep interpolates between parent circuits rather than drawing unrelated masks.

Partial Spider GPTQ eval (Table 15, seed 0) supports a bit-width-dependent dispute optimum: at 3-bit, TaCQ-heavy allocation (λ_{bias}=0) reaches 63.2%, beating the balanced point (λ_{bias}=0.5, 59.9%) and tracking the MSG baseline (64.8%). 2-bit GPTQ evals were interrupted by GPU OOM on parallel rounds; Figure 6 plots the completed 3-bit partial curve with TaCQ / v2 mult dashed baselines (26.0 / 32.6 at 2-bit; 64.8 / 66.2 at 3-bit). This aligns with the mask-overlap story: harsh 2-bit favors routing-centric outliers (§9.1), while milder 3-bit still rewards load-bearing MLP mass.

9.2 Validating causal isolation via targeted suppression

Low mask overlap (§9.1) shows T-DSO selects a different 0.35% outlier set than TaCQ; it does not by itself prove those weights are causally necessary for text-to-SQL. We therefore run a circuit knockout on the full FP16 model: scale masked weights in memory without quantization,

$$W_{\text{scaled}} = W \cdot \big(1 + (\alpha - 1) \cdot M\big), \tag{8}$$

where M=1 on the 24.4M T-DSO outliers (B0 heavy global mask, tdso_b0_heavy_0.35.pt) and M=0 elsewhere. For α=0.8, task weights are multiplied by 0.8 while the remaining 99.65% of parameters stay at unit scale—suppression is routed exclusively through the mask.

Table 16 shows a 29-point collapse in execution accuracy under suppression, with the largest drops on medium/hard/extra splits (hard 54.0→19.0). The same 20% penalty applied only to masked weights leaves general English intact: three short non-SQL probes (capital, sentence completion, definition) remain fluent and grammatical (e.g., "Paris"; full-sentence replies) after suppression. If M tagged random high-magnitude weights, dampening 24.4M parameters would either have little effect or degrade both SQL and English; the observed task-selective failure mode rules out that null.

This knockout is the mechanistic justification for using transcoder-guided saliency to allocate the 0.35% FP16 budget: the selected sub-network behaves like a monosemantic reasoning graph for schema binding and SQL routing, separable from background fluency weights. The boost control (α=1.2) still hurts Spider (-8.2 pt), so linear amplification does not recover FP16 performance—causality is established by suppression, not by claiming in-memory scaling substitutes for mixed-precision PTQ.

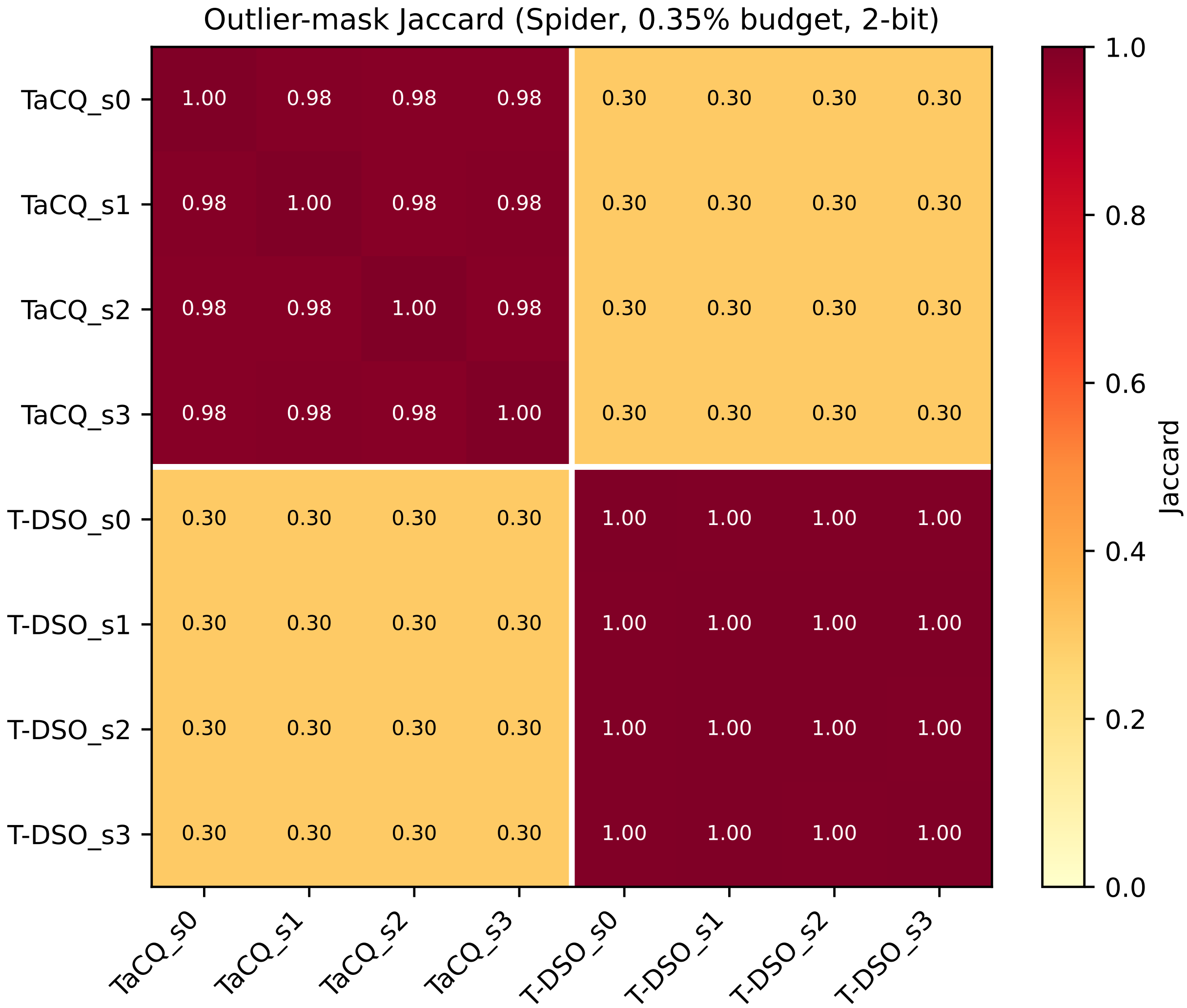

Figure 1: Cross-seed and cross-method Jaccard among 0.35% outlier masks (Spider, 2-bit). Upper-left block: TaCQ MSG seeds 0–3 (J=0.983–0.985). Lower-right block: v2 mult seeds 0–3 (J=0.995–0.997). Off-diagonal blocks: method gap (J≈0.30), not seed noise.

9.3 Contrastive SNR squeeze (leak-free α search)

Beyond one-sided knockout (Eq. 8), we test contrastive in-memory scaling that boosts task-masked weights while dampening background weights within the same tensors: $W \leftarrow W \odot (\alpha_{\text{task}} M + \alpha_{\text{base}}(1-M))$. To avoid calibration leakage on Spider dev (our effective test set), we search $(\alpha_{\text{task}}, \alpha_{\text{base}})$ only on a stratified 150-example train holdout from the 1360-example calibration pool, freeze the winner, then evaluate dev once.

Table 17 shows a verified +1.0 pt dev gain under leak-free train selection, with a contemporaneous FP16 control at 67.8%. Combined with circuit knockout (§9.2), this supports a causal operational hierarchy: the transcoder-guided mask identifies weights whose selective amplification and background dampening improve SQL routing without PTQ, retraining, or extra inference cost. Aggressive background suppression collapses train accuracy to 25.3%, confirming the circuit is tightly balanced. Train-subset accuracies (∼72–73%) are not comparable to dev (different split).

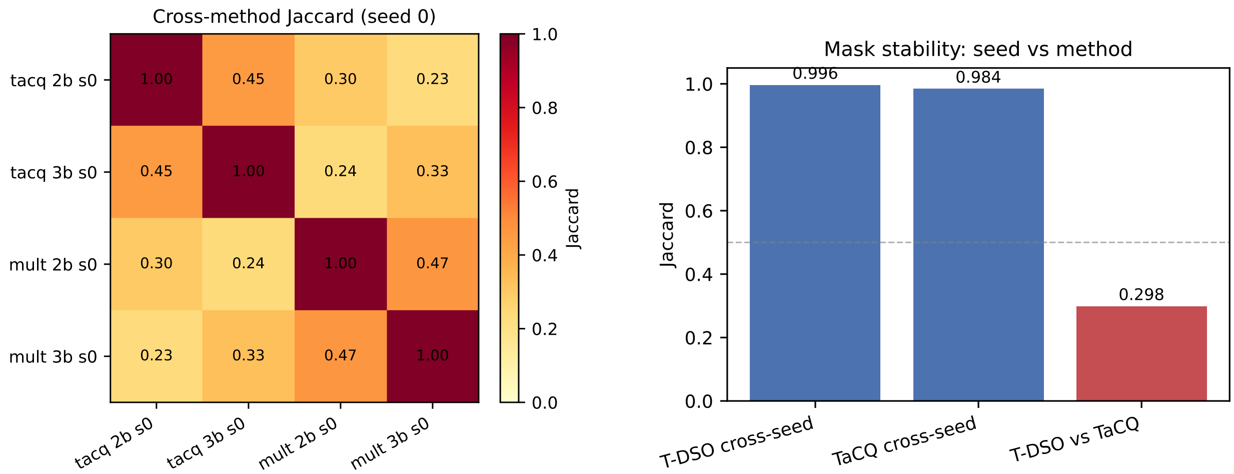

Figure 2: Left: Cross-method Jaccard at seed 0 for 2- and 3-bit masks (J=0.298 / 0.331 on the diagonal off-blocks; intra-method bitwidth overlap >0.95). Right: Mean cross-seed J vs. cross-method J—seed variance is an order of magnitude smaller than the method gap.

Table 12: Outlier-mask overlap at matched 0.35% budget (24.4M weights). $|A \cap B|/|A|$ = fraction of T-DSO (v2 mult) outliers also kept by TaCQ.

Comparison	Jaccard	Shared
T-DSO v2 mult vs. TaCQ 2-bit	0.298	46%
T-DSO v2 mult vs. TaCQ 3-bit	0.331	50%

9.4 Transcoder-native feature audit

Weight-level overlap (§9.1) and knockout (§9.2) validate that T-DSO selects a causally task-relevant subnetwork; here we ask whether the transcoder features that drive extraction look task-specific rather than generic LM patterns. During v2 extraction we accumulate per-layer task-feature density (meta.layer_feature_density in the mask checkpoint); mid layers 4–14 carry the largest fraction of discriminative activations, consistent with the layer-wise weight disagreement in Figure 4. We additionally rank Top-K feature IDs by cumulative clean−corrupt activation Δa on Spider contrastive pairs (scripts/analysis/feature_audit.py) and manually label a subset (Appendix A.3, Table 20). Paper 1 stops at this static audit; runtime feature traces and per-step overlays are deferred to the agentic extension (§11).

9.5 Efficiency and failure modes

Wall-clock. Saliency extraction for T-DSO v2 matches TaCQ asymptotics (one forward–backward per calibration example) plus per-layer transcoder encode/decode. Full 1360-pair Spider extraction ≈ 40 min on 4× H200; [TODO: exact TaCQ timing table].

Failure modes. v2 boost ($g_{\text{CE}} \cdot (1 + \lambda g_{\text{align}})$) degenerates: the 2-bit model rambles (SELECT SELECT ...) and hits the generation cap without EOS, confirming that expanding the TaCQ mask with circuit bonuses is worse than intersecting signals. When transcoder reconstruction is weak (expected layer cosine ∼0.5), v1 align-only masks underperform CE; hybrid mult is robust on average but seed-sensitive (§9.6).

9.6 Seed sensitivity

Both TaCQ and v2 mult exhibit non-trivial seed variance at 2-bit Spider (Table 4). v2 mult exec spans 26.2–32.6% (n=4, mean 29.4%, sample std 2.7 pp); TaCQ spans 25.0–29.6% (mean 26.5%, std 2.0 pp;

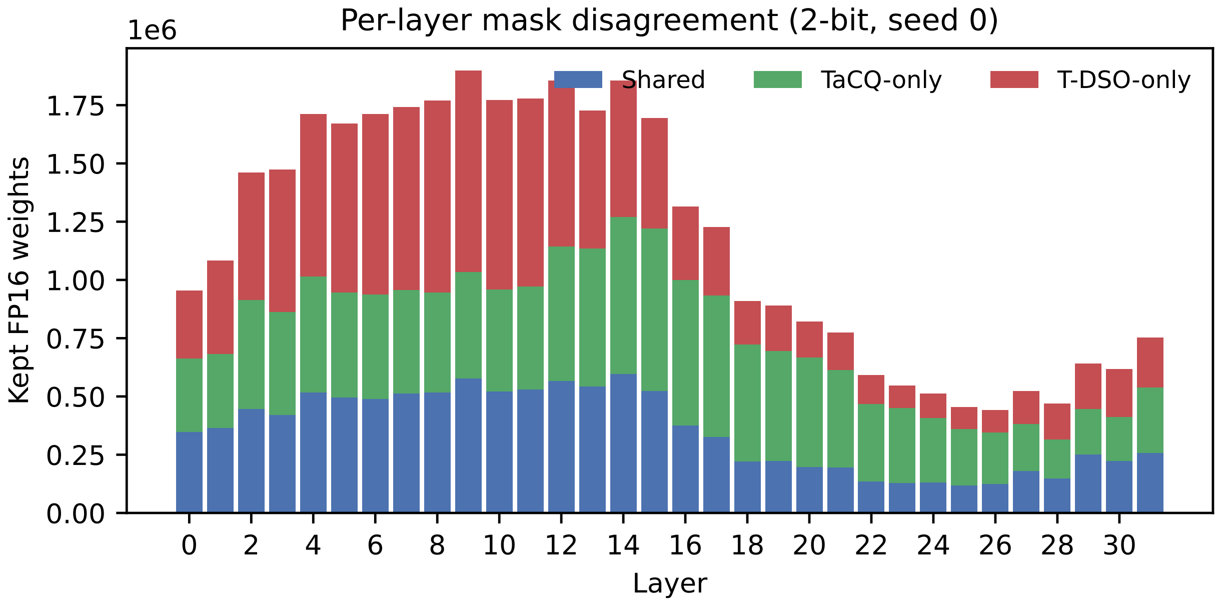

Figure 3: Per-layer kept-weight decomposition (2-bit, seed 0). Mid layers 4–14 carry the largest T-DSO-only and TaCQ-only mass; early/late layers agree more.

Table 13: Per-component kept-weight disagreement (2-bit, seed 0; counts in millions). Inter = shared FP16 outliers; columns show method-exclusive mass.

Component	Inter	TaCQ-only	T-DSO-only
down_proj	2.28	3.54	2.90
gate_proj	2.04	3.76	2.28
up_proj	1.80	2.91	2.49
k_proj	1.56	0.69	1.32
o_proj	1.51	1.04	2.16
q_proj	1.47	0.92	1.30
v_proj	0.55	0.34	0.76
Attention (sum)	5.10	2.99	5.53
MLP (sum)	6.12	10.21	7.67

Table 4).

What seeds control.

- TaCQ (serial_number=s): shuffles the 128-example Spider-train calibration subset (preprocess_calibration_datasets), sets PyTorch RNG for gradient accumulation, and sets GPTQ –seed s. Corrupt ΔW is fixed (seed0 checkpoint) for fair comparison.
- v2 mult (seed s): regenerates contrastive corruptions on all 1360 train pairs (data_prep_contrastive.py) and sets extraction –seed s.

Thus seed variance reflects calibration stochasticity, not different quantizers or budgets.

Seed sensitivity vs. method divergence. Cross-seed Jaccard among v2 mult masks is 0.995–0.997 (Figure 2, Table 18): only ~36–56k of 24.4M kept weights differ per seed pair despite 6.4 pp exec spread. The circuit is therefore stable; small mask perturbations and/or GPTQ noise dominate downstream exec at 2-bit—not

Table 14: Anchored-union mask morph: Jaccard vs. parent masks as λ_{bias} sweeps dispute re-ranking (consensus core locked; seed 0). Log: v3_dispute_sweep.log.

λ_{bias}	2-bit J		3-bit J	
	TaCQ	T-DSO	TaCQ	T-DSO
0.00	0.708	0.460	0.715	0.496
0.25	0.643	0.511	0.653	0.547
0.50	0.574	0.575	0.590	0.606
0.75	0.514	0.640	0.540	0.661
1.00	0.462	0.705	0.490	0.724

Table 15: Dispute sweep Spider exec (%, seed 0; partial — 2-bit OOM on parallel GPTQ). Baselines: TaCQ 64.8 / v2 mult 66.2 at 3-bit.

λ_{bias}	2-bit exec	3-bit exec
0.00 (TaCQ-heavy)	—	63.2
0.50 (balanced)	—	59.9

wholesale circuit replacement. In contrast, v2 vs. TaCQ cross-method J=0.298: methods select different circuits, not different seeds of the same circuit.

TaCQ cross-seed masks. TaCQ MSG mask Jaccard across seeds is 0.983–0.985 (Figure 1; tacq_crossseed_jaccard.log): high but lower than v2's 0.995–0.997, with ∼187–214k kept weights differing per seed pair vs. ∼36–56k for v2. Despite comparable exec spread (2.0 vs. 2.7 pp std), TaCQ's outlier set is therefore more calibration-tie-sensitive at the weight level than CE∩transcoder intersection— consistent with polysemantic CE gradients amplifying small shuffle effects in global top-k.

Reporting. We report mean ± std for both methods in Table 4. Seed 1 v2 (26.2%) ties TaCQ s0 (26.0%), so claims should emphasize paired Δ (mean v2 − mean TaCQ) rather than best-seed cherry-picking.

9.7 Causal vs. load-bearing masks

The B0 (F_{task}) and MSG (contrastive CE) masks occupy different functional roles (Table 6, §8). Causal: B0 identifies weights whose in-memory suppression collapses Spider (−29 pt knockout; §9.2) and whose mild boost/damp squeeze improves FP16 (+1.0 pt). Under matched ΔW corruption without GPTQ re-fit, B0 preserves enough routing structure for v2 mult to beat TaCQ on average (+3.4 pt). Load-bearing: MSG marks weights whose FP16 retention is necessary for survival after GPTQ at 2-bit—26.0% with Spider calibration vs. 0.6% with C4, holding mask fixed. B0 fails this role (0.0% GPTQ with C4; 18.0% under Phase B stacked GPTQ even with Spider calib). Thus transcoder circuit attribution and contrastive CE saliency answer different questions: causal routing vs. quantization-critical mass. Our headline PTQ comparison uses the intersection combiner (v2 mult), which partially bridges both signals.

10 Limitations

- Dictionary coverage (current). Full per-layer CRV transcoders exist only for Llama-3.1-8B (§6.1). Scale extension uses public instruct SAEs (Goodfire 70B) and EleutherAI skip-transcoders where full CRV stacks are unavailable; see paper/SCALE_PLAN.md.
- Approximate MLP targets. TopK transcoders reconstruct MLP output imperfectly; we rely on error-node semantics from the circuit-tracer literature.
- Compute. Transcoder encode/decode adds overhead vs. TaCQ; [TODO: quantify].

Table 16: Circuit knockout on FP16 Llama-3.1-8B (Spider dev exec, 1034 examples). Same mask and 20% perturbation magnitude; only routing via M differs from a global penalty.

Condition	α	Exec (%)
FP16 baseline	1.0	67.8
Suppress (task circuit only)	0.8	38.9 (−28.9 pt)
Boost (control)	1.2	59.6 (−8.2 pt)

Table 17: Contrastive squeeze: train grid (n=150) and single-pass dev (job 2488279). FP16 dev control 67.8% re-scored in the identical harness (job 2489136).

Config	α_{task}	α_{base}	Exec (%)
Train grid (selection only)			
Baseline	1.00	1.00	72.0
Gentle (selected)	1.05	0.98	73.3
Moderate	1.10	0.95	65.3
Aggressive	1.15	0.90	25.3
Dev (n=1034, matched harness)			
FP16 control	1.00	1.00	67.8
Gentle squeeze	1.05	0.98	68.8 (+1.0)

- Seed / calibration variance. Both methods vary with calibration seed (Table 4): v2 mult spans 26.2–32.6% (n=4); TaCQ paired sweep in progress. v2 cross-seed masks are stable ($J\approx0.996$); exec spread arises from ∼50k-weight perturbations at 2-bit, not circuit replacement.
- Task coverage. Primary evidence on Spider; broader task sweep [TODO: in progress].
- Scale evidence (in progress). Submission centers on 8B CRV; Phase 2 (Llama-3.3-70B + Goodfire SAE) is queued per Table 2.

11 Systems Outlook: Dynamic Sparse Overlays

Beyond static PTQ comparison, T-DSO motivates a Dynamic Sparse Overlay execution model not available to TaCQ. We do not claim these systems results in v1; they define the research arc if v1 dissociation holds empirically.

Agentic hot-swapping. TaCQ produces one task-conditioned mask baked into the quantized checkpoint—configuration is fixed for the inference job. If sub-skills (schema linking, SQL generation, self-correction) prefer different outlier sets, TaCQ must compromise with a single global top-k. T-DSO's feature-space objective naturally supports separate calibration streams per sub-skill and, with an orthogonality penalty on mask overlap, non-redundant overlays M_{link}, M_{gen}, M_{fix}. At runtime, only the FP16 outlier values differ; the 2-bit base $W_{2\text{bit}}$ stays resident while overlay tables stream (e.g., via PCIe into on-chip SRAM). [TODO: Formalize QAL + orthogonality penalty; implement mask streaming benchmark.]

KV-cache inheritance. Multi-step text-to-SQL pipelines often chain agents as separate API calls. Each call re-tokenizes history and recomputes attention from scratch—$O(N^2)$ per step. With a shared frozen $W_{2\text{bit}}$ and hot-swapped overlays, a downstream sub-circuit inherits the prior step's KV cache; only overlay weights and the active T-DSO mask change. Time-to-first-token for steps 2+ approaches cache-append cost rather than full prefill. [TODO: Microbenchmark TTFT: static TaCQ vs. overlay swap on two-step Spider prompt.]

Table 18: Cross-seed outlier-mask Jaccard (0.35% budget). v2 mult from crossseed_jaccard.log; TaCQ MSG from tacq_crossseed_jaccard.log.

Comparison	J	Notes
v2 mult s0 vs. s1	0.996	52k weights differ
v2 mult s0 vs. s2	0.995	56k weights differ
v2 mult s0 vs. s3	0.996	55k weights differ
v2 mult s1 vs. s2	0.997	36k weights differ
v2 mult vs. TaCQ (2-bit, s0)	0.298	13.2M unique each
v2 mult vs. TaCQ (3-bit, s0)	0.331	12.3M unique each
TaCQ s0 vs. s1	0.985	187k weights differ
TaCQ s0 vs. s2	0.983	210k weights differ
TaCQ s0 vs. s3	0.984	194k weights differ

Positioning for reviewers. Dimensions 2–3 are enabled by feature dissociation plus mixed-precision structure; v1 validates Dimension 1 (Spider exec accuracy at matched budget). Systems claims require separate engineering evidence and belong in an extended version or companion systems section once built.

12 Conclusion

Low-bit PTQ forces a choice about which tiny fraction of weights must remain in full precision. We contribute three reviewer-facing takeaways on Llama-3.1-8B Spider (full detail: REVIEWER_PROOF_TAKEAWAYS.md).

Double dissociation. TaCQ MSG and T-DSO B0 select largely disjoint 0.35% outlier sets (J=0.30). Knockout and contrastive squeeze show B0 tracks causal routing weights; MSG paired with Spider calibration preserves load-bearing structure for 2-bit survival (Table 11).

2-bit domain shift. Native gptq.llama controls show C4 calibration collapses Spider 2-bit exec (MSG: 26.0% $\rightarrow$ 0.6% when calib corpus changes), exposing a benchmarking confound at extreme bitwidth (§8).

Multi-regime B0. The B0 mask wins under matched methods: +3.4 points mean (+6.6 best seed) at 2-bit with Spider ΔW, and a verified +1.0 point FP16 gain via leak-free contrastive squeeze (Table 17). The v3 dispute sweep further shows bit-width-dependent optimal dispute allocation: 3-bit exec peaks at TaCQ-heavy λ_{bias}=0 (63.2%) while mask Jaccard crosses from MSG- to T-DSO-dominated near λ_{bias}=0.5 (Tables 14–15).

TaCQ answers outlier selection with contrastive CE gradients; intersecting CE with transcoder task-circuit attribution selects a different circuit and improves 2-bit Spider execution on average at matched budget. T-DSO preserves the TaCQ quantization stack for fair comparison.

Scale (committed). We extend T-DSO to transcoder-compatible models beyond 8B (Table 2): EleutherAI skip-transcoders (1B ladder), Goodfire instruct SAEs (8B ablation, 70B scale headline), pending full-layer CRV releases. TaCQ-only baselines on Qwen2.5-7B / Llama-3-70B provide cross-architecture context without dictionaries. Runbook: paper/SCALE_PLAN.md.

Future work: multi-task masks, bootstrap CI on paired Δ, full-layer 70B CRV when available.

Reproducibility. Code, masks, and replication logs: [TODO: anonymous repo URL for submission].

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A Appendix

A.1 Hyperparameters

Table 19: T-DSO / TaCQ shared hyperparameters.

Parameter	Value
Outlier fraction	0.35%
Calibration pairs	128
Max sequence length	[TODO: 2048?]
Feature threshold quantile	0.95
Transcoder TopK	128
GPTQ	true-sequential, 128 samples
Ranking	top_p_sparse

A.2 Contrastive corruption schema

[TODO: Describe Spider / GSM8K / HumanEval corruption types from data_prep_contrastive.py.]

A.3 Transcoder-native feature audit

Layer feature density. Each v2 mask checkpoint stores per-layer counts of task-discriminative transcoder activations (meta.layer_feature_density). Summarize with:

```
python scripts/analysis/layer_feature_density.py \
  --mask masks/tdso_v2_mult.pt \
  --out tacq_data/results/spider_layer_feature_density.json
```

(Runs on CPU; no GPU needed.)

Top-K feature ranking. Rank features by cumulative $\Delta a = \mathrm{ReLU}(a^{\mathrm{clean}} - a^{\mathrm{corr}})$ on Spider contrastive pairs (128 examples, 95th-percentile threshold per example, matching §5):

```
sbatch scripts/sbatch_feature_audit_debug.sh
```

Uses debug partition (–account=a135), 1 GPU, 45 min wall time. For a full run on normal, set MAX_PAIRS=1280 and increase time. Outputs:

```
tacq_data/results/spider_layer_feature_density.json
tacq_data/results/spider_feature_audit.json
```

Output JSON lists Top-K feature_id per layer with cum_delta; fill Table 20 by manual inspection (schema / JOIN / aggregation / filter / syntax / other). Optional: cross-reference CRV auto-interp labels when available.

Promoted tokens and activating contexts. For the global Top-20 features from the audit JSON, run:

```
sbatch scripts/sbatch_feature_interpret_debug.sh
```

This writes tacq_data/results/spider_feature_interpretations.json with (i) top vocabulary tokens promoted by each feature's decoder row × lm_head and (ii) max-activating token snippets on Spider clean calibration text. Use both signals to assign categories in Table 20.

Attribution-graph-style cards (circuit-tracer format). For panels matching Anthropic's Attribution Graphs "Understanding and Labeling Features" (quantile activating examples with per-token coloring, top/bottom logits, histogram), run:

sbatch scripts/sbatch_feature_cards_debug.sh

Open tacq_data/results/spider_feature_cards.html in a browser. Paper 2 extends this ladder to per-overlay features and runtime agent traces (paper/PAPER2_AGENTIC_OVERLAY.md).

Table 20: Manual audit template for Top task-discriminative transcoder features (Spider). Populate from spider_feature_audit.json; example rows are placeholders.

Layer	Feature ID	$\sum \Delta a$	Category	Notes
[TODO: L7]	[TODO: fid]	[TODO: –]	schema / JOIN / agg / filter / syntax / other	[TODO: trigger token or span]
[TODO: L11]	[TODO: fid]	[TODO: –]	[TODO: –]	[TODO: –]
[TODO: L14]	[TODO: fid]	[TODO: –]	[TODO: –]	[TODO: –]

A.4 Additional results

[TODO: Full Spider difficulty breakdown; 3-bit table; ablation RTN vs GPTQ ΔW.]

A.5 Transcoder reconstruction verification

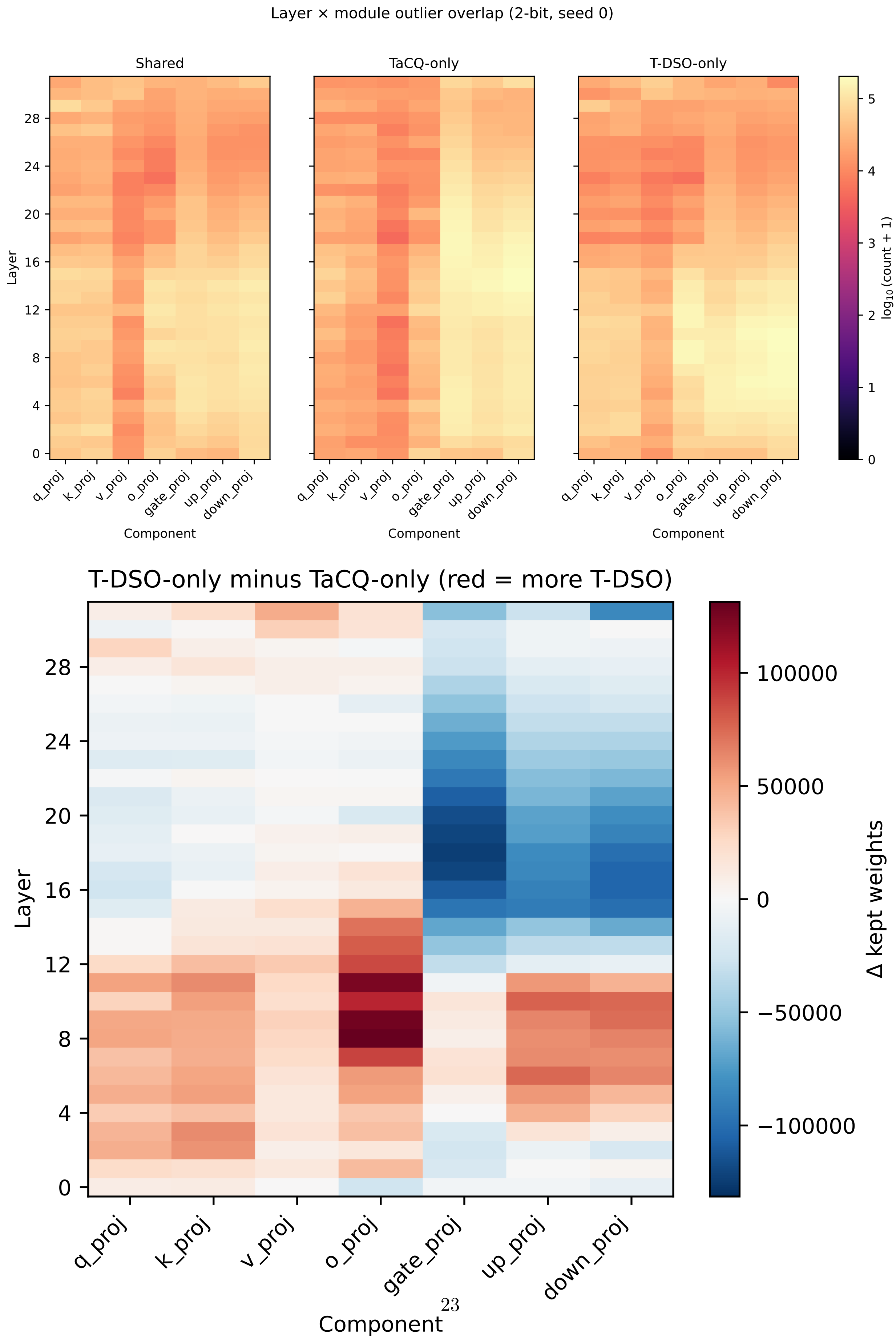
[TODO: Reference authors_recon_check.py; cosine per layer ≈ 0.5 expected.]

A.6 Algorithm pseudocode

[TODO: Algorithm 1: T-DSO saliency extraction.]

A.7 Broader impact

[TODO: Standard LLM compression + interpretability dual-use paragraph for ICLR checklist.]


Layer × module outlier overlap (2-bit, seed 0)
Shared
TaCQ-only
T-DSO-only
Layer
Component
q_proj
k_proj
v_proj
o_proj
gate_proj
up_proj
down_proj
$\log_{10}$(count + 1)
T-DSO-only minus TaCQ-only (red = more T-DSO)
Δ kept weights

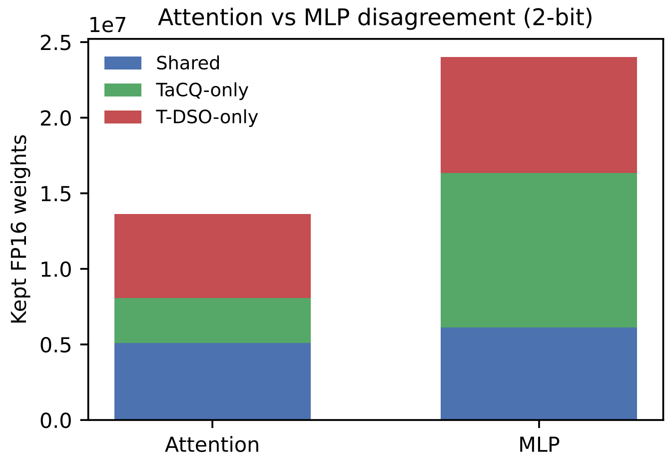


Figure 5: Attention vs. MLP disagreement (2-bit, seed 0). T-DSO retains 5.5M attention-only outliers vs. 3.0M TaCQ-only; MLP shows the reverse (10.2M vs. 7.7M), consistent with routing-vs-memory dissociation.

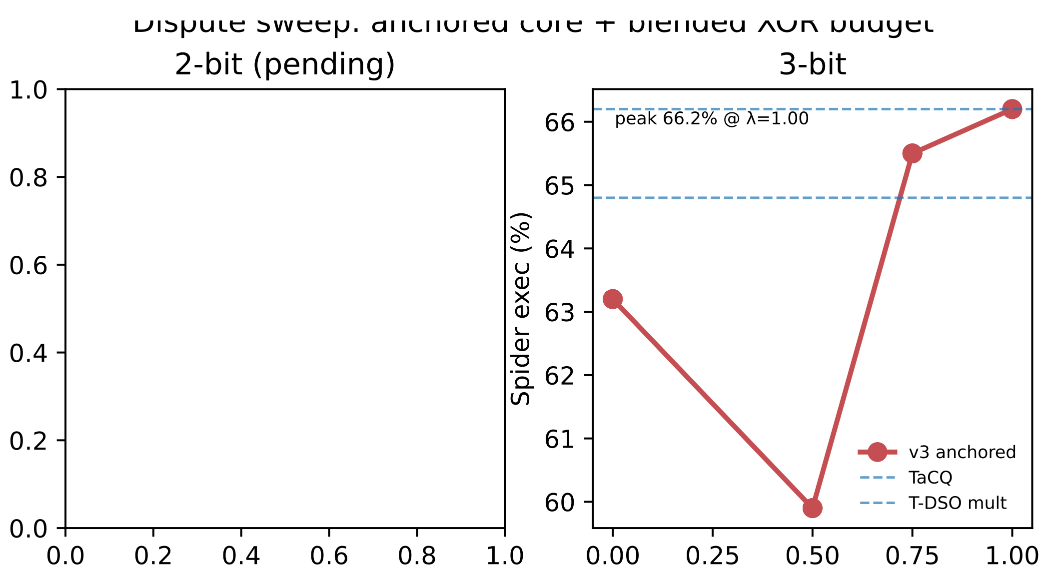


Figure 6: Dispute sweep (anchored core + blended XOR budget, seed 0). Partial results: 3-bit only (λ=0 peaks at 63.2% vs. 59.9% at λ=0.5); 2-bit GPTQ pending. Dashed lines: TaCQ and v2 mult baselines.